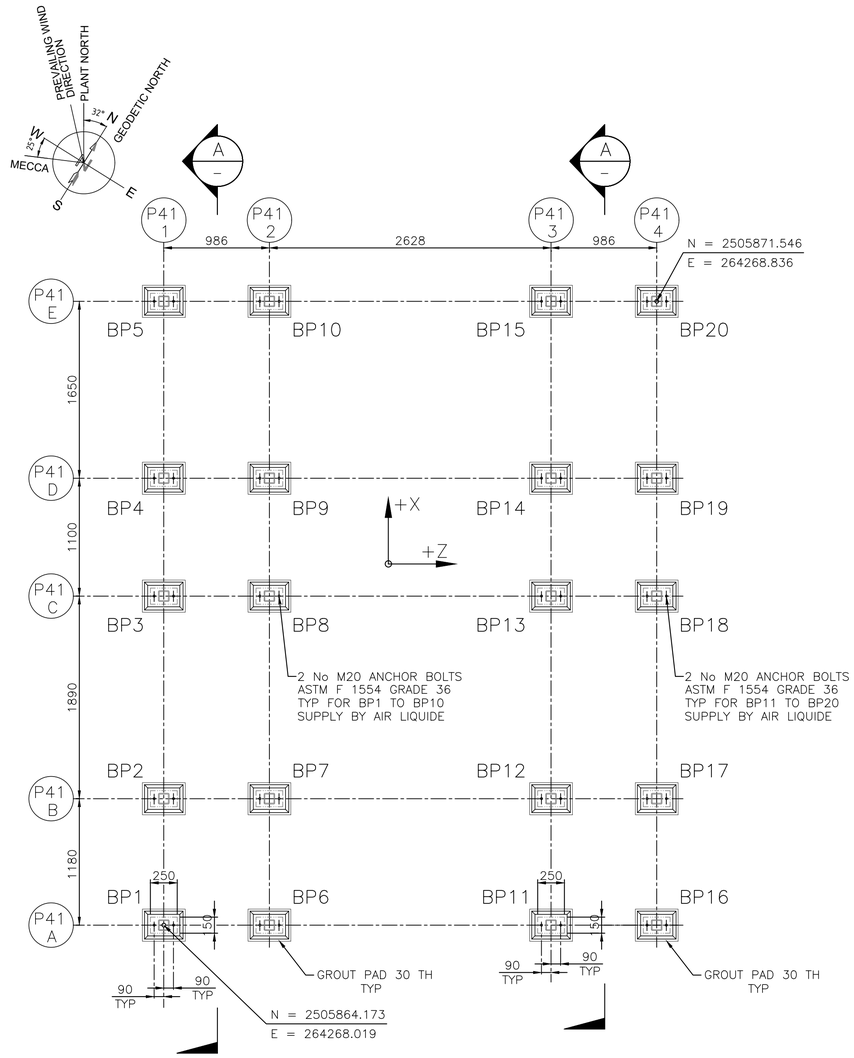
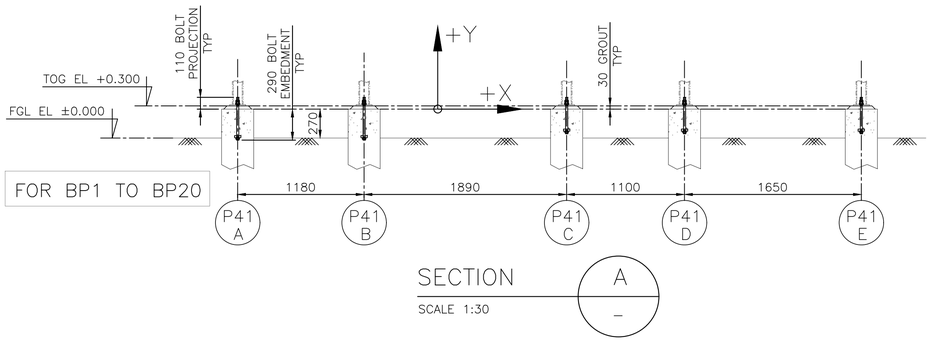
DRAWING NUMBER: MD - 216 - 3980 - EG - CV - GAD - V551-1491 | REV: D01



FOUNDATION PLAN LOX BACK UP PUMP CASING

SCALE 1:30



SECTION A–A

SCALE 1:30

FOUNDATION LOAD : LOX BACK UP PUMP CASING

SL No	Description	BP1			BP2			BP3			BP4			BP5			Remarks
		X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	
1	DEAD LOAD	0.0	-1.1	0.0	-0.2	-3.6	0.0	0.2	-5.4	0.0	-0.7	-5.0	0.0	0.7	-2.5	0.0	A. ALL LOADS FURNISHED IN THIS TABLE ARE UNFACTORED ACTING LOADS.
2	EQUIPMENT LOAD	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	-0.4	-0.8	0.0	0.3	-1.4	0.2	
3	PIPE LOAD	0.0	0.3	0.0	-0.3	-3.2	0.0	0.3	-3.7	0.0	0.0	-2.0	0.0	0.0	0.0	0.0	
4	LIVE LOAD	0.3	-0.5	0.2	-0.4	-4.1	0.0	0.7	-4.7	-0.2	-1.1	-4.7	-0.3	0.9	-3.1	0.0	B. PIPE LOAD AND EQUIPMENT LOAD SHALL BE CONSIDERED UNDER DEAD LOAD CATEGORY.
5	PIPE LOAD +X	0.4	0.5	0.0	0.6	0.0	0.0	0.7	0.2	0.0	0.5	0.3	0.0	0.7	-0.7	0.0	
6	PIPE LOAD +Z	0.0	-0.2	0.0	0.0	-0.2	0.0	0.0	-0.5	-0.3	-0.2	-1.8	-1.1	0.0	-0.6	-0.3	
7	WIND LOAD +X	0.7	1.0	0.2	1.3	0.0	0.0	1.3	0.2	0.0	1.9	0.6	0.2	1.5	-2.1	0.2	C. FOR REVERSE DIRECTION OF LOAD CASE 5 TO 10, MULTIPLY ALL VALUES WITH -1.
8	WIND LOAD +Z	0.2	2.4	1.1	0.2	4.1	2.0	-0.4	5.9	2.2	0.6	7.9	3.2	-0.5	3.8	1.9	
9	SEISMIC LOAD +X	0.2	0.3	0.0	0.3	0.0	0.0	0.3	0.0	0.0	0.4	0.2	0.0	0.3	-0.5	0.0	
10	SEISMIC LOAD +Z	0.0	0.4	0.3	0.0	0.6	0.3	0.0	0.6	0.3	0.0	0.7	0.4	0.0	0.5	0.2	

SL No	Description	BP6			BP7			BP8			BP9			BP10			Remarks
		X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	
1	DEAD LOAD	-0.2	-5.7	-0.6	-0.2	-8.5	-0.6	0.4	-5.0	0.0	-0.9	-5.2	0.0	0.8	-3.6	0.0	A. ALL LOADS FURNISHED IN THIS TABLE ARE UNFACTORED ACTING LOADS.
2	EQUIPMENT LOAD	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	-0.4	-0.8	0.0	0.3	-1.5	-0.2	
3	PIPE LOAD	0.0	-2.9	-0.3	-0.3	-6.2	-0.3	0.3	-3.7	0.0	0.0	-2.0	0.0	0.0	0.0	0.0	
4	LIVE LOAD	0.2	-6.8	-0.9	-0.7	-17.0	-1.3	1.3	-15.1	-0.2	-2.7	-16.3	-0.3	1.8	-10.4	-0.4	B. PIPE LOAD AND EQUIPMENT LOAD SHALL BE CONSIDERED UNDER DEAD LOAD CATEGORY.
5	PIPE LOAD +X	0.5	0.7	0.0	0.6	-0.3	0.0	0.7	0.0	0.0	0.5	0.3	0.0	0.7	-0.8	0.0	
6	PIPE LOAD +Z	0.0	0.0	0.0	0.0	0.2	-0.2	0.0	0.5	-0.3	0.0	1.7	-1.1	-0.2	0.6	-0.3	
7	WIND LOAD +X	1.1	2.0	0.2	1.6	-0.8	-0.2	1.7	0.3	0.0	2.8	1.0	0.0	2.4	-2.4	-0.2	C. FOR REVERSE DIRECTION OF LOAD CASE 5 TO 10, MULTIPLY ALL VALUES WITH -1.
8	WIND LOAD +Z	0.2	-0.7	1.9	-0.2	-2.0	2.7	0.4	-5.9	2.2	-0.3	-7.7	3.3	0.5	-3.9	1.9	
9	SEISMIC LOAD +X	0.3	0.4	0.0	0.3	-0.2	0.0	0.3	0.0	0.0	0.5	0.3	0.0	0.4	-0.4	0.0	
10	SEISMIC LOAD +Z	0.0	-0.2	0.4	0.0	-0.3	0.4	0.0	-0.6	0.3	0.0	-0.7	0.4	0.0	-0.5	0.3	

SL No	Description	BP11			BP12			BP13			BP14			BP15			Remarks
		X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	
1	DEAD LOAD	-0.2	-5.8	0.7	-0.2	-8.6	0.6	0.4	-5.1	0.0	-0.9	-5.2	0.0	0.8	-3.6	0.0	A. ALL LOADS FURNISHED IN THIS TABLE ARE UNFACTORED ACTING LOADS.
2	EQUIPMENT LOAD	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	-0.4	-0.8	0.0	0.3	-1.4	0.2	
3	PIPE LOAD	0.0	-4.0	0.4	-0.3	-7.4	0.4	0.3	-3.7	0.0	0.0	-2.0	0.0	0.0	0.0	0.0	
4	LIVE LOAD	0.2	-6.8	0.9	-0.7	-17.0	1.3	1.3	-15.1	0.2	-2.7	-16.3	0.3	1.8	-10.4	0.4	B. PIPE LOAD AND EQUIPMENT LOAD SHALL BE CONSIDERED UNDER DEAD LOAD CATEGORY.
5	PIPE LOAD +X	0.5	0.7	0.0	0.6	-0.3	0.0	0.7	0.0	0.0	0.5	0.3	0.0	0.7	-0.8	0.0	
6	PIPE LOAD +Z	0.0	0.0	0.0	0.0	-0.2	-0.2	0.0	-0.5	-0.3	0.0	-1.7	-1.1	0.2	-0.6	-0.3	
7	WIND LOAD +X	1.1	2.0	-0.2	1.5	-0.8	0.2	1.6	0.3	0.0	2.7	0.9	0.0	2.2	-2.2	0.0	C. FOR REVERSE DIRECTION OF LOAD CASE 5 TO 10, MULTIPLY ALL VALUES WITH -1.
8	WIND LOAD +Z	-0.2	0.5	1.9	0.0	2.1	2.7	-0.5	5.9	2.2	0.3	7.7	3.2	-0.8	4.1	1.9	
9	SEISMIC LOAD +X	0.3	0.4	0.0	0.3	-0.2	0.0	0.3	0.0	0.0	0.5	0.3	0.0	0.4	-0.5	0.0	
10	SEISMIC LOAD +Z	0.0	0.2	0.4	0.0	0.3	0.4	0.0	0.6	0.3	0.0	0.7	0.4	0.0	0.5	0.3	

SL No	Description	BP16			BP17			BP18			BP19			BP20			Remarks
		X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	
1	DEAD LOAD	0.0	-1.1	0.0	-0.2	-3.6	0.0	0.2	-5.4	0.0	-0.7	-5.0	0.0	0.7	-2.5	0.0	A. ALL LOADS FURNISHED IN THIS TABLE ARE UNFACTORED ACTING LOADS.
2	EQUIPMENT LOAD	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	-0.4	-0.8	0.0	0.3	-1.5	-0.2	
3	PIPE LOAD	0.0	0.5	-0.2	-0.3	-3.1	0.0	0.3	-3.7	0.0	0.0	-2.0	0.0	0.0	0.0	0.0	
4	LIVE LOAD	0.3	-0.5	-0.2	-0.4	-4.1	0.0	0.7	-4.7	0.2	-1.3	-4.8	0.3	0.9	-3.2	0.0	B. PIPE LOAD AND EQUIPMENT LOAD SHALL BE CONSIDERED UNDER DEAD LOAD CATEGORY.
5	PIPE LOAD +X	0.4	0.5	0.0	0.6	0.0	0.0	0.7	0.2	0.0	0.5	0.3	0.0	0.7	-0.7	0.0	
6	PIPE LOAD +Z	0.0	0.2	0.0	0.0	0.2	0.0	0.0	0.5	-0.3	0.2	1.8	-1.1	0.0	0.6	-0.3	
7	WIND LOAD +X	0.7	1.0	-0.2	1.1	0.0	0.0	1.1	0.0	0.0	1.8	0.6	-0.2	1.5	-2.0	-0.2	C. FOR REVERSE DIRECTION OF LOAD CASE 5 TO 10, MULTIPLY ALL VALUES WITH -1.
8	WIND LOAD +Z	0.0	-2.5	1.1	-0.4	-4.2	2.0	0.2	-5.9	2.2	-0.6	-7.9	3.2	0.3	-3.6	1.9	
9	SEISMIC LOAD +X	0.2	0.3	0.0	0.3	0.0	0.0	0.3	0.0	0.0	0.4	0.2	0.0	0.3	-0.5	0.0	
10	SEISMIC LOAD +Z	0.0	-0.4	0.3	0.0	-0.6	0.3	0.0	-0.6	0.3	0.0	-0.7	0.4	0.0	-0.5	0.2	

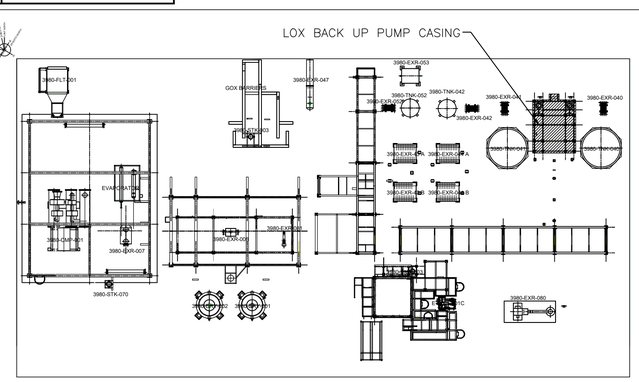
NOTE: PIPE LOAD +X & PIPE LOAD+Z ARE THE HORIZONTAL LOADS GENERATED DUE TO PIPE STRESS. THESE TWO LOADS SHALL BE CONSIDERED ACTING SIMULTANEOUSLY IN OPERATING CONDITION.

NOTES

1. ALL DIMENSIONS ARE IN MILLIMETER.
2. ALL LEVELS ARE IN METER.
3. FGL LEVEL ±0.0 CORRESPONDS TO RL 912.350.
4. THE LOADS INDICATED IN LOAD DATA TABLE ARE ACTUAL. FOUNDATION DESIGNER SHOULD CONSIDER NECESSARY MARGIN.
5. SIZE OF FOUNDATION AND PEDESTAL SHALL BE DECIDED BY CIVIL DESIGN ENGINEER AS PER GIVEN LOADING.
6. ADDITIONAL REBAR REQUIREMENT AROUND ANCHOR BOLTS TO ENSURE NO FAILURE OF CONCRETE SHALL BE DECIDED BY CIVIL DESIGN ENGINEER AS PER GIVEN LOADING.
7. ALL GROUT SHALL BE NON SHRINK CEMENTITIOUS GROUT UNLESS NOTED OTHERWISE.

LEGEND :

CL – CENTER LINE
CG – CENTER OF GRAVITY
CONC – CONCRETE
EL – ELEVATION
EQP – EQUIPMENT
FGL – FINISHED GROUND LEVEL
Gr – GRADE
MPa – MEGA PASCAL
mm – MILLIMETER
No – NUMBER / NUMBERS
SQ – SQUARE
TH – THICKNESS
TOC – TOP OF CONCRETE
TOG – TOP OF GROUT
TOP – TOP OF SHIM PLATE
TYP – TYPICAL



KEY PLAN

REV	DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	17/02/2023 AS-BUILT	--	STY/SYC/SHH	--	BDA/PRI/JPD	
C01	10/08/2020 ISSUED FOR CONSTRUCTION	--	STY/SYC/SHH	--	BDA/PRI/JPD	
A01	08/06/2020 ISSUED FOR INFORMATION	--	STY/SYC/SHH	--	BDA/PRI/JPD	

DOCUMENT NUMBER | REFERENCE DOCUMENTS

Air Liquide ENGINEERING & CONSTRUCTION

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DRAWING NUMBER				REVISION	STATUS
12126-60-01-CS-151131				0	AFD
DATE	03/06/2020	DATE	04/06/2020	DATE	05/06/2020
DRAWN	SGH	CHECKED	ABM	APPROVED	DS

LARSEN & TOUBRO DRAWING NUMBER: 019045_080_C_V_24225_018 | REVISION: 2

OUTOTEC DRAWING NUMBER: 900707 __GAU01 __AG22__MTL01 __39827 | DOCUMENT ID: OU100680052 | REVISION: 2

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DRAWN	--	--	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	STY/ SYC/ SHH	06/06/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	--	--	OXYGEN PLANT
APPROVED	BDA/ PRI/ JPD	08/06/2020	FOUNDATION LOAD - LOX BACK UP PUMP CASING
EMPLOYER			
SCALE	1:30	A1	DRAWING NUMBER: MD - 216 - 3980 - EG - CV - GAD - V551-1491 | REV: D01